Predicting potentially hazardous asteroids with a neural network

Deep Learning project report

Name: Akshat Chauhan

1. Introduction

This project builds a binary classifier that predicts whether an asteroid is a potentially hazardous asteroid (PHA) from its orbital and physical measurements. The data is the Kaggle Asteroid Dataset (sakhawat18), which compiles records from NASA's Jet Propulsion Laboratory. Each row is one asteroid, and the target column `pha` holds Y or N. We use only the numeric columns as inputs.

The full table has 45 columns, 33 of them numeric. After dropping rows with no `pha` label, roughly 938,600 labelled asteroids remain. Hazardous asteroids make up only about 0.2% of them, so most of the work in this project concerns metrics and imbalance. A model that never flags anything would score above 99% accuracy, so we track recall, precision, and AUC instead, and use class weights and a tuned decision threshold to deal with the imbalance. We also test dropout and batch normalization separately before combining them.

Because we have such a small percentage of hazardous asteroids, our accuracy metric is not a proper metric for determining our model's worth, since it will always have a high value. Instead, the cost of missing a potentially hazardous asteroid is far more costly, so we will focus on maximizing recall. Additionally, since NASA has very strong definitions for defining potentially hazardous asteroids, we expect our recall value to be quite high already.

2. Data and preprocessing

The data was downloaded with `kagglehub`. We kept the 33 `float64` columns as features. Histograms of a 10,000-row sample show that many of these features are heavily right-skewed. Many describe the size or makeup of the asteroid, so the skew is plausible: large, massive asteroids are rare and small ones are common.

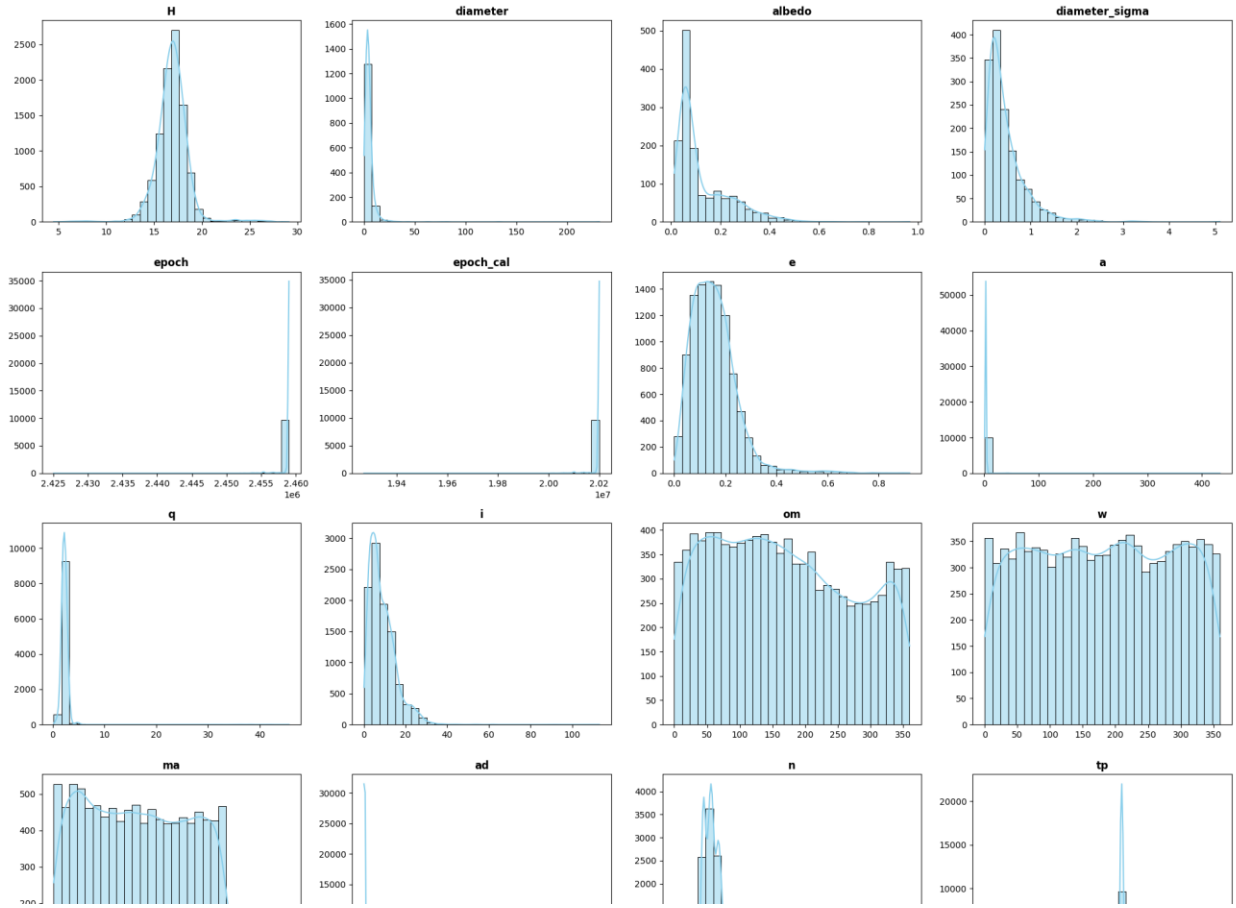


Figure 1. Histograms for 16 of the 33 numeric features (10,000-row random sample).

Preprocessing was done with a scikit-learn pipeline:

- Rows with a missing `pha` value were dropped, and the label was mapped from Y/N to 1/0.
- Missing feature values were filled with the column median, and every feature was standardized with `StandardScaler`.
- The data was split with stratification into 70% training, 15% validation, and 15% test (random seed 42). The preprocessor was fit on the training split only.

Batch normalization cannot rescale the raw inputs to the first layer, which is why we standardize the inputs up front and use `BatchNormalization` only between hidden layers.

Split	Share	Rows	Hazardous rows
Training	70%	about 657,000	n/a
Validation	15%	140,790	310
Test	15%	140,791	310

Table 1. Data splits. Training counts are derived from the other two splits.

3. Baseline model

The baseline is a plain feed-forward network with no regularization: a 256-unit ReLU layer, a 64-unit ReLU layer, and a single sigmoid output. It was compiled with Adam and binary cross-entropy, tracking binary accuracy, precision, recall, and AUC. We trained it for 50 epochs with a batch size of 512. The final training loss was 0.0014, the validation loss was 0.0021, and the validation AUC was 0.9869.

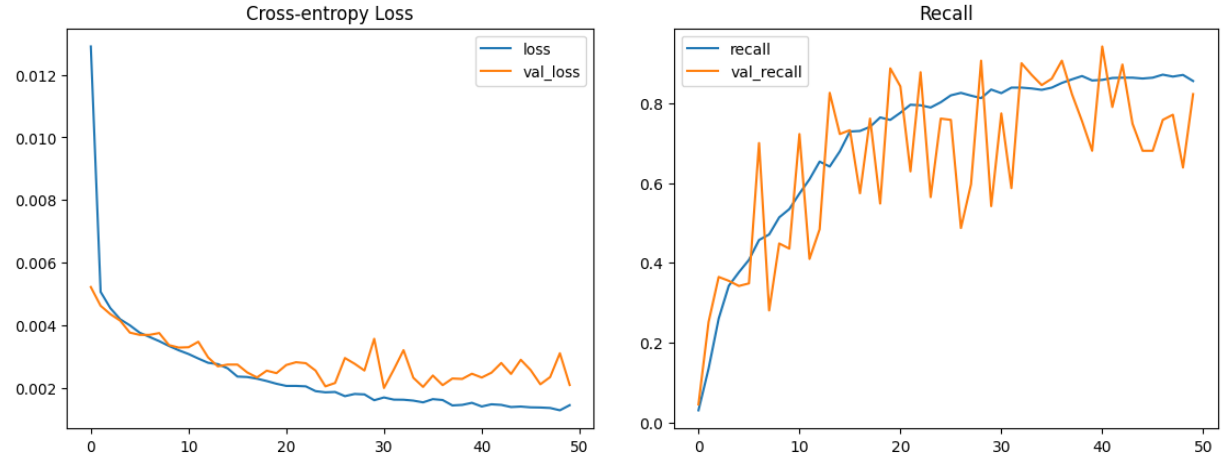


Figure 2. Baseline model: cross-entropy loss (left) and recall (right) per epoch.

Around epochs 15 to 20, training loss keeps falling while validation loss starts to bounce upward. This is a textbook sign of overfitting and is what we expected with no regularization. We address it with early stopping and dropout.

The validation curves also oscillate widely. The validation set holds only 310 hazardous asteroids out of 140,790 rows (about 0.2%), so catching or missing a handful of them swings recall sharply from one epoch to the next. For the same reason accuracy tells us little: with so few positives it stays very high regardless. Recall on the hazardous class is the more meaningful measure, both mathematically and for the purpose of the task.

4. Regularization and tuning

4.1 Early stopping and learning rate search

We added early stopping (patience 10, minimum improvement 0.001, best weights restored). We then used Keras Tuner with random search to look for a learning rate between 1e-4 and 1e-2, sampled on a log scale. The search ran 5 trials and maximized validation AUC. The best learning rate was about 0.000175.

With the tuned learning rate, validation loss rose from 0.0021 to 0.0036, but validation AUC rose from 0.9869 to 0.9990. We select AUC because it summarizes how well the model separates the two classes across every threshold. The ROC curve plots recall against the false positive

rate, so a larger area means the model can reach high recall at a lower false positive rate. At the default 0.5 threshold, the tuned model had a validation precision of 0.64 and a recall of 0.53.

4.2 Dropout and batch normalization separately

Before combining the two techniques, we trained one model with dropout and one with batch normalization so that we could see what each does on its own. The dropout model applies a rate of 0.5 after each hidden layer. The batch normalization model adds a `BatchNormalization` layer after each hidden dense layer. Both use the tuned learning rate and early stopping.

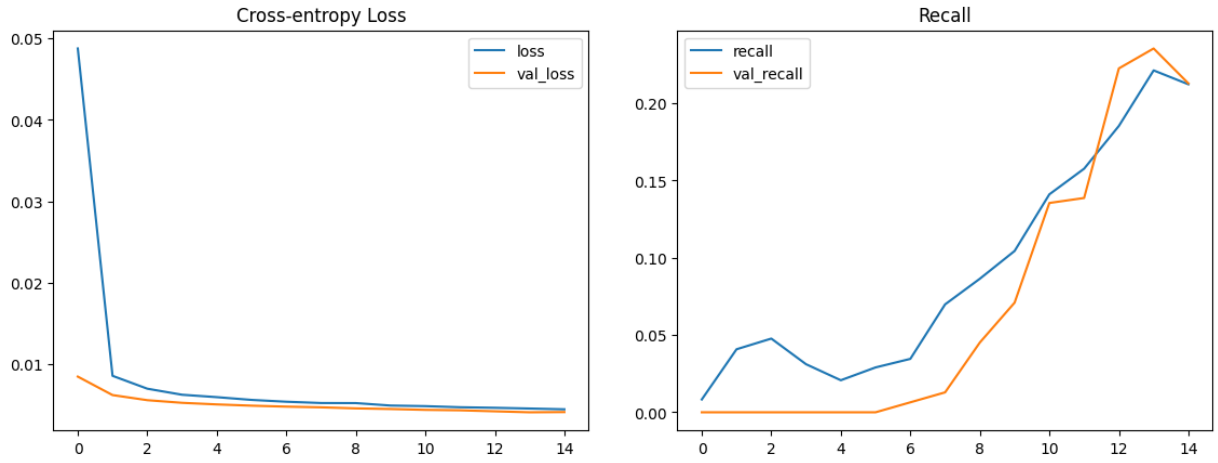


Figure 3. Dropout model (rate 0.5): loss (left) and recall (right).

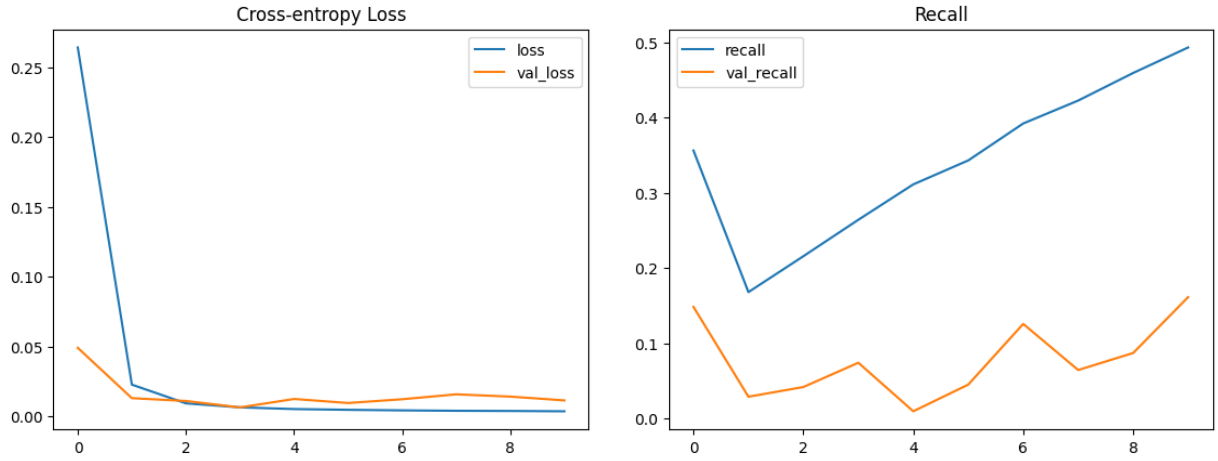


Figure 4. Batch normalization model: loss (left) and recall (right).

Model	Val. loss	Val. AUC	Val. recall
Baseline	0.0021	0.9869	0.8226
Dropout (0.5)	0.0041	0.9969	0.2129
Batch normalization	0.0115	0.9975	0.1613

Table 2. Validation metrics from the last epoch of each training history.

Dropout narrowed the gap between training and validation loss, and batch normalization reduced the wild oscillation in the validation curves. Dropout also was able to make our graphs smoother. Both lowered recall at the default threshold, which we attribute again to the class imbalance. That is a problem we're going to fix.

5. Final model with class weights

To address the drop in recall, we combined both techniques, lowered the dropout rate to 0.2, and trained with balanced class weights computed from the training labels. Class weights make errors on the rare hazardous class count for more in the loss.

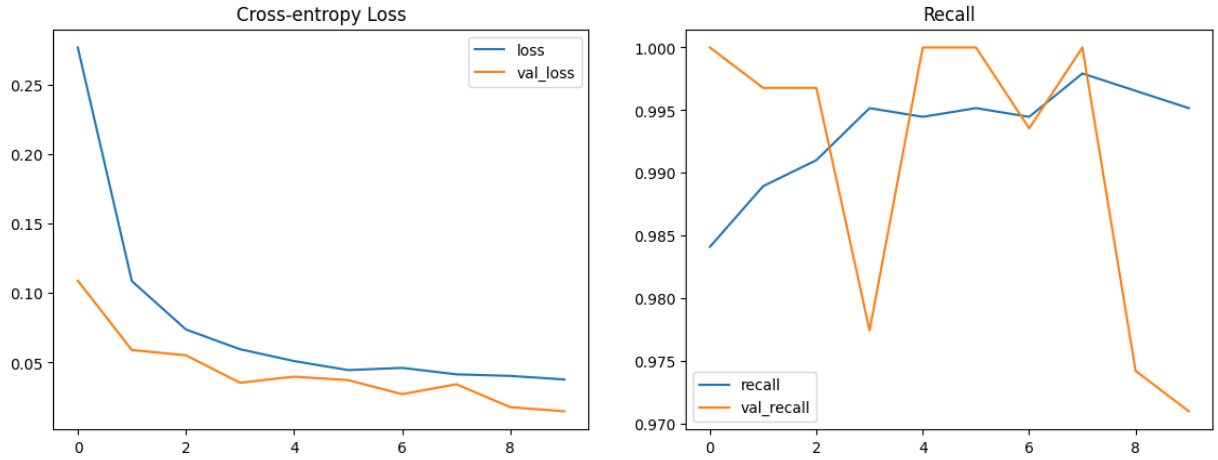


Figure 5. Final model (batch normalization, dropout 0.2, class weights): loss (left) and recall (right).

Model	Val. loss	Val. AUC	Val. recall
Baseline	0.0021	0.9869	0.8226
Final model	0.0149	0.9978	0.9710

Table 3. Baseline versus final model, last-epoch validation metrics. Validation precision fell from 0.83 to 0.29.

The final model raised AUC and recall, but at the default 0.5 threshold it flags far too many asteroids. On the validation set it produced 1,880 false positives and no false negatives, which gives a precision of 0.14 and a recall of 1.00. Because the class weights push the model toward flagging, it is playing it safe, and the 0.5 cutoff is too low for it.

6. Choosing a decision threshold

To find a better cutoff, we plotted the precision-recall curve on the validation set and chose the threshold that maximizes F1. That threshold is 0.8044, which gives a validation precision of 0.2331 and a recall of 0.8323.

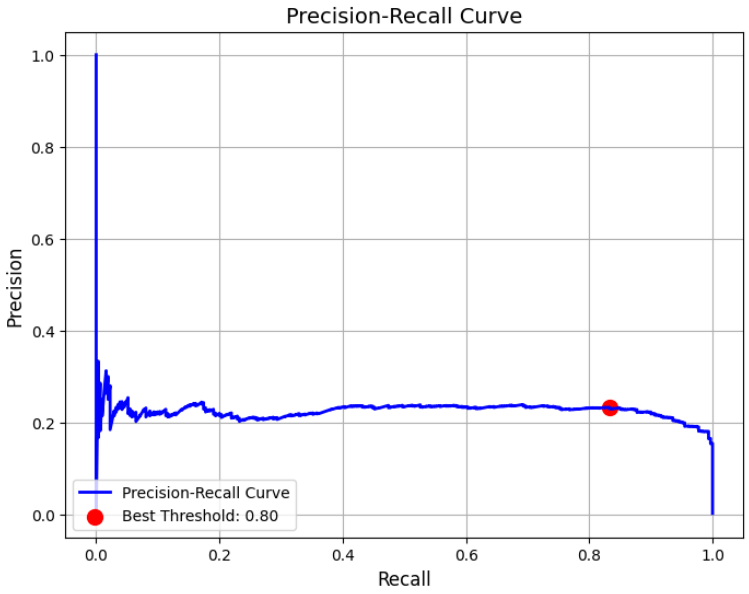


Figure 6. Precision-recall curve on the validation set, with the F1-maximizing threshold marked in red.

Threshold	Precision	Recall	F1	Confusion matrix (TN, FP / FN, TP)
0.5	0.14	1.00	0.25	138,600 1,880 / 0 310
0.8044	0.23	0.83	0.36	139,631 849 / 52 258

Table 4. Validation results for the hazardous class at two thresholds.

Moving the threshold costs some recall in exchange for a large drop in false positives, from 1,880 to 849.

Again, since the proportion of PHAs is so small in our dataset, we will always value recall a lot, but a low precision is still a grave issue so that is why we chose to address it.

7. Test set results

We evaluated the final model on the held-out test set using the threshold chosen on validation data. It caught 242 of 310 hazardous asteroids (78%) with 927 false positives. Precision and recall are close to the validation figures (about 0.23 and 0.83), which suggests the model did not overfit, as the regularization was meant to prevent.

Data	Precision	Recall	F1	Confusion matrix (TN, FP / FN, TP)
Validation	0.23	0.83	0.36	139,631 849 / 52 258
Test	0.21	0.78	0.33	139,554 927 / 68 242

Table 5. Hazardous-class results at threshold 0.8044.

8. Comparison with NASA's definition

NASA classifies an asteroid as potentially hazardous mainly from two quantities: an absolute magnitude H of 22 or less, and a minimum orbit intersection distance (MOID) of 0.05 or less. We checked the asteroids our model flagged against that rule. The model flagged 1,169 asteroids in the test set, and 1,147 of them satisfy both conditions, an agreement of 98.12%.

That figure measures how many flagged asteroids fit the rule. It does not measure how many asteroids that fit the rule were flagged. Also, only 242 of the 1,169 flagged asteroids carry a hazardous label in the dataset, so at least 905 of the 927 false positives satisfy the H and MOID conditions as well.

9. Conclusion and limitations

The final network reached a test recall of 0.78 at a precision of 0.21, and 98.12% of the asteroids it flagged meet NASA's H and MOID criteria. The project was a useful exercise in handling a severely imbalanced target: accuracy was uninformative, class weights and threshold selection did most of the work, and the regularization methods behaved differently when tested one at a time.

A neural network is not the best model for this dataset. NASA's definition is a pair of logical threshold rules on two features, and a decision tree can represent rules like that directly. We did not train a tree in this notebook, so how closely it would match the labels is untested.

Appendix: model configurations

Model	Hidden layers	Regularization	Learning rate	Class weights
Baseline	Dense 256, Dense 64	None	Adam default	No
Tuned baseline	Dense 256, Dense 64	Early stopping	0.000175	No
Dropout	Dense 256, Dense 64	Dropout 0.5 x2	0.000175	No
Batch norm	Dense 256, Dense 64	BatchNorm x2	0.000175	No
Final	Dense 256, Dense 64	BatchNorm and dropout 0.2, x2	0.000175	Balanced

Table 6. All models use ReLU hidden layers, a sigmoid output, batch size 512, and up to 50 epochs. Models other than the baseline use early stopping.

References

Sakhawat, Asteroid Dataset. Kaggle. https://www.kaggle.com/datasets/sakhawat18/asteroid-dataset

[Add any other sources you cite, such as NASA's definition of a potentially hazardous asteroid.]